1. Perform the following tasks using git-bash
    a. Initialize a Git repository for a data science project
    b. Push it to GitHub to enable version control
    c. Create a python file name as program1.py and write a python program to print multiplication table of any number
    d. Commit the code and push the changes and check the changes on remote repository(GitHub)
    e. Create a new branch named GIT-EXAM and create new file named as program2 and write a program to find greatest of two numbers
    f. Commit the changes and push them and demonstrate the difference of changes in master and GIT-EXAM branch
2. Perform the following tasks using git-bash
    a. Initialize a Git repository for a data science project
    b. Push it to GitHub to enable version control
    c. Create a python file name as program1.py and write a python program to check even odd number.
    d. Commit the code and push the changes and check the changes on remote repository(GitHub)
    e. Create a new branch named GIT-EXAM and create new file named as program2 and write a program to find the reverse of a string
    f. Commit the changes and push them and demonstrate the difference of changes in master/main and GIT-EXAM branch
    g. Create a pull request to merge the changes from GIT-EXAM branch to master/main branch
3. Perform the following tasks using git-bash
    a. Initialize a Git repository for a data science project
    b. Push it to GitHub to enable version control
    c. Create a new branch named GIT-EXAM-1 and create new file named as program2 and write a program to check if given string is palindrome
    d. Create a python file name as program1.py and write a python program to print squares of numbers from 1 to 10
    e. Commit the code and push the changes and check the changes on remote repository(GitHub)
    f. Create a new branch named GIT-EXAM-2
    g. Pull the changes from branch GIT-EXAM-1
    h. create new file named as program2.py and write a program to check if given string is palindrome
    i. Modify the code in program1.py(to demonstrate conflicts)

    j. Commit the changes and push them and demonstrate the difference of changes in master/main, GIT-EXAM-1 and GIT-EXAM-2 branch
    k. Create a pull request to merge the changes from GIT-EXAM-2 branch to GIT-EXAM-1 branch
4. Perform the following tasks using git-bash
    a. Initialize a Git repository for a data science project
    b. Push it to GitHub to enable version control
    c. Create a new branch named GIT-EXAM-1 and create new file named as program2 and write a program to check if given string is palindrome
    d. Create a python file name as program1.py and write a python program to print squares of numbers from 1 to 10
    e. Commit the code and push the changes and check the changes on remote repository (GitHub)
    f. Create a directory named as IGNORE and add test1.txt, test2.txt files into it
    g. Create .gitignore file and add the IGNORE directory into it to ignore the changes made to it
    h. Commit and push the code and demonstrate that IGNORE directory and its files are not getting pushed on remote
5. Perform the following tasks using Git-Hub Desktop
    a. Initialize a Git repository for a data science project
    b. Push it to GitHub to enable version control
    c. Create a new branch named GIT-EXAM-1 and create new file named as program2 and write a program to check if given string is palindrome
    d. Create a python file name as program1.py and write a python program to print squares of numbers from 1 to 10
    e. Commit the code and push the changes and check the changes on remote repository (GitHub)
    f. Create a directory named as IGNORE and add test1.txt, test2.txt files into it
    g. Create .gitignore file and add the IGNORE directory into it to ignore the changes made to it
    h. Commit and push the code and demonstrate that IGNORE directory and its files are not getting pushed on remote
6. Using Git-bash Desktop Perform the following tasks
    a. Clone repository from the path - https://github.com/KalyanM45/AI-Project-Gallery.git
    b. Demonstrate that branch creation is disallowed for this repository
    c. Fork this AI project repository
    d. Create a feature branch (GIT-EXAM) to implement changes

   e. Change/Add your code in the branch
   f. Submit a pull request for review.
   g. During the review process, resolve any merge conflicts and incorporate feedback to ensure smooth integration and collaboration.
7. Using Git-Hub Desktop Perform the following tasks
   a. Clone repository from the path - https://github.com/KalyanM45/Spam-Email-Detection.git
   b. Demonstrate that branch creation is disallowed for this repository
   c. Fork this AI project repository
   d. Create a feature branch (GIT-EXAM) to implement changes
   e. Change/Add your code in the branch
   f. Submit a pull request for review.
   g. During the review process, resolve any merge conflicts and incorporate feedback to ensure smooth integration and collaboration
8. Jenkins
   a. Create a repository on GitHub and upload project files.
   b. Install and launch Jenkins (`http://localhost:8080`).
   c. Install required plugins (Git, GitHub, Pipeline).
   d. Create a new Pipeline job in Jenkins.
   e. Configure **Pipeline script from SCM** and add GitHub repository URL.
   f. Add a `Jenkinsfile` to the repository.
   g. Configure build trigger (Poll SCM or GitHub Webhook).
   h. Push code changes to GitHub.
   i. Verify that Jenkins automatically triggers the build.

9. Docker
   a. Create a project folder and add AI model files (app.py, requirements.txt).
   b. Create a Dockerfile in the project directory.
   c. Write instructions in the Dockerfile to set up environment and run the model.
   d. Build Docker image using Docker
   e. Run the Docker container
   f. Verify container is running
   g. Test the application in browser or using API (localhost:5000).
   h. Stop the container after testing
10. Utilize Docker Compose to orchestrate and manage multiple containers for a web-based AI application, including separate containers for the AI model, database, and frontend services.
11. Set up a local Kubernetes cluster with Mini Kube to deploy and scale your Dockized AI application.

12. Set up a CI/CD pipeline with Jenkins, Docker, and Kubernetes to automate the development-to deployment process for your AI model.